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Sock orthotic

Abstract

A knit sock orthotic is provided in which a pattern of padding is knit into the sock to create an interface between the foot and the shoe or weight-bearing surface to correct an abnormal biomechanical condition of the foot, or to relieve pressure on focal areas of structural overload of the foot, or both.

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Background/Summary

CROSS-REFERENCE TO RELATED CASES (1) This application claims priority to U.S. Provisional Patent Application No. 63/474,333, entitled “Sock Orthotic,” filed on Aug. 9, 2023, which is hereby incorporated by reference.

BACKGROUND

- (1) The human foot is so well designed and engineered that when it functions in an ideal or neutral position, with the subtalar joint neutral and the midtarsal joint locked, it requires little or no muscle activity to support us. Unfortunately, many feet cannot and do not function in their optimal positions due to compensation for biomechanical abnormalities that are inherited and expressed.
- (2) For most purposes, an “ideal” foot is one which, in its neutral position, functions with the back of the heel bone relatively vertical while the horizontal level of the forefoot bones rests on the flat ground, with each of the five metatarsal heads bearing its fair share of load. This relatively perpendicular relationship between the horizontal forefoot and the vertical rearfoot, in neutral position, is largely determined during the growing years due to developmental torsion of the talus bone in the ankle.
- (3) If the talus does not torque enough, the forefoot will be inverted or higher on the inside, relative to the rearfoot. Obviously, one cannot function with the inside of the forefoot above the ground, so the rearfoot subtalar joint compensates to lower the arch by pronation of the subtalar joint. If the talus torques too much, the inside edge of the forefoot would be lower than the outside, requiring the rearfoot subtalar joint to compensate by supination, or rolling outwards. Deviations on either side of “ideal” result in a foot that functions sub-optimally, requiring additional muscular support and subjecting it to overuse symptoms. In any case, compensation of the rearfoot subtalar joint will occur by necessity to position the forefoot horizontally on the flat ground surface.
- (4) In addition to biomechanical compensation symptoms, deviations away from “ideal” may result in abnormally high focal areas of pressure from a metatarsal that may be structurally too low or long, or from a joint that may be excessively mobile so that it does not assume its fair share of weight distribution. Over time, these areas are subjected to mechanical overload resulting in subsequent inflammation and pain.
- (5) Fortunately, all these problems are well understood, and treatment modalities exist to compensate for biomechanical abnormalities. Successful intervention typically requires that therapeutic efforts be directed to the causative conditions which, if properly addressed, result in resolution of symptoms.
- (6) The range of currently available biomechanical therapy includes application of tape support, wedging to compensate for excessive pronation or supination, and padding to alleviate areas of focal pressure overload. If basic therapy is effective, the problem may be solved with wedge or pad placement in shoes or with over-the-counter supports. If more sophisticated therapy is necessary, a prescription orthotic may be designed and fabricated to better control abnormal biomechanical compensation and/or accommodate for areas of pressure overload. Such pads may be known by a number of names including, e.g.: cushion, correction, orthotic pad, shoe insert pad, biomechanical pad, biomechanical correction, and corrective pad.
- (7) However, such solutions are often unsatisfactory, creating a significant gap in patient compliance because they are shoe specific and occupy space within footwear. Not all shoes tolerate orthotics and orthotics take up space in shoes. Furthermore, existing cushioned socks only provide a general increase in softness.

(8) Thus, what is needed is an orthotic device that remains with the foot, provides a therapeutic interface for the foot, and is easily transferrable.

Description

BRIEF DESCRIPTION OF THE DRAWINGS

- (1) The embodiments are illustrated by way of example and not limitation in the accompanying drawings, in which like references indicate similar elements, and in which.
- (2) FIG. 1 is a top view of an embodiment of a sock orthotic;
- (3) FIG. 2 is a bottom view of an embodiment of a sock orthotic with a cobra pad configuration,
- (4) FIG. 3 is a bottom view of an embodiment of a sock orthotic with a metatarsal pad configuration;
- (5) FIG. 4 is a bottom view of an embodiment of a sock orthotic with a dancer's pad configuration;
- (6) FIG. 5A is an upper front left perspective view of an embodiment of a sock orthotic for the left foot with a metatarsal pad configuration;
- (7) FIG. 5B is an upper front left perspective view of an embodiment of a sock orthotic for the right foot with a metatarsal pad configuration;
- (8) FIG. 6A is an upper front left perspective view of an embodiment of a sock orthotic for the left foot with a cobra pad configuration;
- (9) FIG. 6B is an upper front left perspective view of an embodiment of a sock orthotic for the right foot with a cobra pad configuration;
- (10) FIG. 7A is an upper front left perspective view of an embodiment of a sock orthotic for the left foot with a dancer's pad configuration;
- (11) FIG. 7B is an upper front left perspective view of an embodiment of a sock orthotic for the right foot with a dancer's pad configuration,
- (12) FIG. 8A is a side view illustrating aspects of an embodiment of a sock orthotic;
- (13) FIG. 8B is a side view illustrating aspects of an embodiment of a sock orthotic;
- (14) FIG. 9 includes top views illustrating biomechanical pad configurations of the prior art;
- (15) FIG. 10 includes perspectives views illustrating biomechanical pad configurations of the prior art;
- (16) FIG. 11 includes top views illustrating biomechanical pad configurations of the prior art; and
- (17) FIG. 12 includes top views illustrating biomechanical pad configurations of the prior art.

DETAILED DESCRIPTION

(18) Embodiments described within disclose a sock orthotic that fills a gap in existing therapeutic modalities by providing a specific biomechanical interface between the foot and the shoe or the foot and the ground. In embodiments, the sock orthotic utilizes specific pad configurations having differential areas of thickness that are intrinsically knit into the pattern of the sole of the sock, creating: 1) wedging corrections to provide biomechanical control and/or 2) focal build-ups to reduce pressure overload on adjacent anatomical structures.

(19) Embodiments of a sock orthotic incorporate biomechanical orthotic corrections into a sock manufactured utilizing pad configurations having varying degrees of intrinsically woven thicknesses and patterns within the bottom contour of the sock. By wearing such embodiments, a user may reduce or eliminate specific lower extremity symptoms, benefitting from an interface that effectively controls, compensates for, or accommodates abnormal biomechanical function or uneven structural pressure distribution in the foot. In addition, embodiments may be available over-the-counter and may be recommended by health-care practitioners or selected by the users based on their presenting symptoms.

(20) Thus, embodiments vary greatly from existing socks that only have areas of uniform thickness for padding, or that rely on elastic compression for support (e.g., such as Archtek socks).

(21) FIG. 1 is a top view of an embodiment of a sock orthotic **100** on a left foot **10**. In FIG. 1, sock **100** includes a sock upper **110** with an opening **112**. Sock upper **110** is shown partially transparent to reveal the position of foot **10**, for reference. Foot **10** includes phalanges **14a . . . 14e**, joints **16a . . . 16e**, and an attached ankle **12**. Sock upper **110**, of FIG. 1 is common to all embodiments. In embodiments, right sock orthotics and left sock orthotics that have the same type of orthotic padding are mirror images of each other and the description of one applies equally to the other.

(22) FIG. 2 is a bottom view of an embodiment **100a** of a sock orthotic including a pad section **102a** with a cobra pad configuration, and with the sock rendered partially transparent to reveal foot **10** within. In FIG. 2, which illustrates the right-side version, sock orthotic **100a** includes a sole **120a**. Sole **120a** includes a base sole section **104a**, a transition section **106a**, and a padded section that, in this embodiment is cobra pad section **102a**. Base sole section **104a** includes a knit and a thickness that serves as a baseline for the sole thickness. Transition section **106a** indicates a section in which the thickness of the sock transitions from the thickness of sole section **104a**, to the increased thickness of the padded section, i.e., cobra pad section **102a**. Cobra pad section **102a** is roughly “J” shaped such that it curls about the perimeter of the heel and extends forward along the arch toward the great toe joint **16a**. FIG. 2 shows sock **100a** from the bottom and the increase in thickness of cobra pad section **102a** extends upward toward the foot, or into the page. A cobra pad is designed to reduce central plantar heel pain and provide support for the arch, for the purpose of alleviating the symptoms of plantar fasciitis, heel spurs and arch pain.

(23) FIG. 2 illustrates, as do FIG. 3-FIG. 7B, how the embodiment of the sock orthotic, when properly worn, positions and retains the orthotic pad section at the proper placement with respect to the foot. These figures also illustrate that the orthotic pad section, being integrated into the sock, travels with the sock and, therefore, may be easily transferred to different footwear with the position of the orthotic unchanged with respect to the foot. Also, the sock orthotic may be used without footwear, with the sock creating an interface between the foot and the floor.

(24) For convenience, in embodiments, each separate embodiment will be labeled **100n**, each sole will be labeled **120n**, each padded section will be labeled **102n**, each base sole section will be labeled **104n**, and each transition section will be labeled **106n**, where “n” is a letter and common for a particular embodiment and its elements.

(25) FIG. 3 is a bottom view of an embodiment **100b** of a sock orthotic including a pad section **102b** with a metatarsal pad configuration. Sock orthotic **100b** is rendered partially transparent to reveal foot **10** within. In FIG. 3, which illustrates the right-side version, sock orthotic **100b** includes a sole **120b**. Sole **120b** includes a base sole section **104b**, a transition section **106b**, and a padded section that, in this embodiment is metatarsal pad section **102a**. Base sole section **104b** includes a knit and a thickness that serves as a baseline for the sole thickness (as do all base sole sections **104n**). Transition section **106b** indicates a section in which the thickness of the sock transitions from the thickness of sole section **104b**, to the increased thickness of the padded section, i.e., metatarsal pad section **102b** (as do all transition sections **106n**). Metatarsal pad section **102b** is roughly triangularly shaped such that it extends from behind joints **16a . . . 16d** to the mid-arch. FIG. 3 shows sock **100b** from the bottom and the increase in thickness of metatarsal pad section **102b** extends upward toward the foot, or into the page, as do all pads **102n** in FIG. 2-FIG. 8B. However, in embodiments, a pad may be knit such that it extends downward from the sock sole and away from the foot. Similarly, in embodiments, a pad may be knit such that the increase in thickness extends both upward toward the foot and downward away from the foot. A metatarsal pad is designed to reduce focal pressure on the metatarsal heads by providing a step-up in thickness from the arch to just behind the metatarsal heads; utilized to alleviate metatarsalgia, capsulitis and neuroma symptoms.

(26) FIG. 4 is a bottom view of an embodiment **100c** of a sock orthotic including a pad section **102c** with a dancer's pad configuration. Sock orthotic **100c** is rendered partially transparent to reveal foot **10** within. In FIG. 3, which illustrates the right-side version, sock orthotic **100c** includes

a sole **120c** Sole **120c** includes a base sole section **104c**, a transition section **106c**, and a padded section that, in this embodiment is dancer's pad section **102c**. Dancer's pad section **102c** is roughly fist shaped such that it lies beneath joints **16b . . . 16d** but not joint **16a**. Dancer's pad section **102c** extends generally to the mid-arch. A dancer's pad is designed to reduce focal pressure beneath the great toe joint **16a** by providing a step-up in thickness adjacent to the symptomatic area; utilized to alleviate sesamoiditis and pain under the great toe joint.

(27) FIG. 5A and FIG. 5B are upper front left perspective views of left-foot and right-foot versions, respectively, of sock orthotic **100b**. FIG. 5A and FIG. 5B further illustrate the shape of metatarsal pad section **102b** and its location with respect to foot **10**.

(28) FIG. 6A and FIG. 6B are upper front left perspective views of left-foot and right-foot versions, respectively, of sock orthotic **100a**. FIG. 6A and FIG. 6B further illustrate the shape of cobra pad section **102a** and its location with respect to foot **10**.

(29) FIG. 7A and FIG. 7B are upper front left perspective views of left-foot and right-foot versions, respectively, of sock orthotic **100c**. FIG. 7A and FIG. 7B further illustrate the shape of dancer's **102c** and its location with respect to foot **10**.

(30) FIG. 8A is a side view illustrating aspects of an embodiment of a sock orthotic. In FIG. 8A, base sock section **104** has a base thickness **108** and padded section **102** has an extra padded thickness **114** (or “accommodation thickness”). Extra padded thickness **114** is added by increasing the thickness of the knitting of sole **120**. Thus, extra padded section **102** is integrated into the weave of sole **120**. Extra padded thickness **114** may represent the thickness of any of the embodiments of padded section **102n**. In some embodiments, extra padded thickness is at least 0.125”, which is considered to be a minimum effective thickness. However, in embodiments, extra padded thickness **114** may be varied as required across the shape of the pad. In embodiments, the non-therapeutic area base sock thickness **108** may be comparable to a traditional athletic sock. In an embodiment, an extra padded section **102** may include a subsection, a pattern pad **118**, that has an increased density compared to the remainder of padded section **102**. In an embodiment, extra padded thickness **114** may represent a thickness when measured under load, i.e., extra padded thickness **114** may be intended to be, e.g., 0.125” when bearing the user's weight, but is configured such that it is greater than 0.125” when unloaded and 0.125” when compressed with the user's anticipated weight. For example, in an embodiment a correlation may be made between a sock size and an estimated user weight. Also, in an embodiment, a plurality of socks for a particular shoe size may be provided, with each sock in the plurality configured to compress to a desired pad thickness for a user within a specific weight range.

(31) FIG. 8B is a side view illustrating aspects of an embodiment of a sock orthotic. In FIG. 8B, base sock section **104** has a base thickness **108** and a transition section **106** that ramps up to extra padded thickness **114** over a distance **122**. As in FIG. 8A, extra padded thickness **114** is added by gradually increasing the thickness of the knitting of sole **120**. Thus, transition section **102** is integrated into the knit of sole **120**. Extra padded thickness **114** may represent the thickness of any of the embodiments of transition section **106n**. As shown in FIG. 2-FIG. 7B, distance **122** may vary between the various transition sections, and even within the same transition section. Thus, the rate at which the thickness of transition section **106** increases may vary as needed to create a desired transition profile. Similarly, the rate at which the thickness of transition section **106** increases may also vary with the result being that transition sections may have non-linear profiles that may include, e.g.: abrupt step changes, gradually increasing slopes, abruptly increase slopes that taper to a flat, and gradually increasing slopes that change to gradually levelling-off slopes.

(32) As shown in FIG. 8B, the total thickness of a sole **120** may be adjusted as needed, e.g., from a base thickness **108** to the sum of base thickness **108** and extra padded thickness **114**. Thus, in some embodiments, the thicknesses of transition sections **106n** and padded sections **102n** may be arbitrarily changed as needed. This may include configurations in which the thickness of the pad may be less than base thickness **108**, such as “cut-out” areas designed to reduce focal

concentrations of pressure. In some such configurations, the thickness of the biomechanical pad may range from a thickness that is less than base thickness **108** and increase to a substantially greater thickness, e.g., to provide a biomechanical pad with an effective pad “cut-out” to offload excessive weight-bearing stress in the symptomatic area.

(33) As illustrated in FIG. **8A** and FIG. **8B**, extra padded section **102** and transition section **106** are thicknesses added to sock sole **120**. These sections are integrated into sole **120** with the effect that the area of extra padding, e.g., transition section **106** and the section of sole **104** over the distance **122**, is unified, i.e., sections **106** and **104** are not distinct in this area, though the area may vary in thickness and density. In embodiments, biomechanical pad corrections may extend upward along the inside or outside edge to provide support and compensation for excessive pronation or supination. For example, such a pad configuration may be found in a Carlton saddle (a soft medial flange **913**) (FIG. **9**). Generally, it should be understood that embodiments may include any biomechanical pad configuration.

(34) FIG. **9** includes top views illustrating biomechanical pad configurations of the prior art. FIG. **9** illustrates the shapes of pads that may be knit into an embodiment of a sock orthotic. Such pads include: a heel lift **901**, a heel spur pad **902**, a heel cut-out **903**, a metatarsal pad **904**, a metatarsal bar **905**, a dancer's pad **906**, a neuroma pad **907**, a toe crest **908**, a scaphoid pad **909**, a soft medial flange **910**, a Morton's extension **911**, a reverse Morton's extension **912**, a Carlton saddle **913**, and a metatarsal cut-out **914**.

(35) FIG. **10** includes perspective views illustrating biomechanical pad configurations of the prior art. FIG. **10** further illustrates the shapes of pads that may be knit into an embodiment of a sock orthotic. Such pads include: first ray cut-out **1001**, first metatarsal cut-out (not shown), an arch pad **1003**, heel pad **901**, heel spur pad **902**, metatarsal bar **905**, metatarsal pad **904**, lateral flange **1004**, medial flange **1005**, Morton's extension **911**, reverse Morton's extension **912**, neuroma pad **907**, and a heel cut-out **1007**.

(36) FIG. **11** includes top views illustrating biomechanical pad configurations of the prior art. FIG. **11** further illustrates the shapes of pads that may be knit into an embodiment of a sock orthotic. Such pads include: heel pad **901**, metatarsal pad **904**, reverse Morton's frame-extension to toes **1101**, reverse Morton's frame-extension to sulcus **1102**, a metatarsal balance pad **1107**, and a pad cut-out **1108**.

(37) FIG. **12** includes top views illustrating biomechanical pad configurations of the prior art. FIG. **12** further illustrates the shapes of pads that may be knit into an embodiment of a sock orthotic. Such pads include: a cut-out pad **1202**, a metatarsal bar pad **1203**, dancer's pad **906**, toe crest pad **908**, neuroma pad **907**, a cuboid offload **1204**, heel spur pad, **902**, Carlton saddle **913**, and a Valgus onlay pad **1205**.

(38) Thus, in embodiments, a sock orthotic may be used to treat a specific condition of the foot by including a knit section in the shape of any of the pad or insert corrections of the prior art associated with the treatment of the specific condition. For example, the following biomechanical correction configurations may be incorporated into an embodiment of a sock orthotic to correct the condition listed: Varus Wedge: used to correct excessive pronation or flatfoot and treat plantar fasciitis and shin splints, Valgus Wedge: used to correct excessive supination and treat high arch foot, Cobra Pad: used to provide arch support, reduce focal heel pressure, and treat heel spur and plantar fasciitis, Metatarsal Pad: used to relieve focal pressure on areas of the forefoot and treat metatarsalgia; Neuroma Pad: used to relieve metatarsal pressure on the nerve and treat interdigital neuroma, Dancer's Pad: used to relieve pressure under the great toe joint and treat sesamoiditis or turf toe; Heel Lift: used to elevate heel and treat Achilles tendonitis or leg length inequality.

(39) In some embodiments, a sock orthotic unit may include some combination of: a right sock orthotic, a left sock orthotic, and a neutral or uncorrected sock. In some embodiments, a sock orthotic may include only a right sock orthotic or a left sock orthotic. In some embodiments, a sock orthotic may be appropriate for either right or left foot.

(40) In an embodiment, extra padded section **102**, e.g., sections **102n**, may be integrated into sock sole **120** during the knitting process by pulling in additional loops of yarn. These additional loops may vary in length, gauge, and choice of material to achieve the desired properties of the therapeutic extra padded section **102**, such as, e.g., a desired thickness, or density. Longer loops may be referred to as “terry.” The additional loops may be integrated in combinations of lengths and/or elastic yarns, such as Lycra, to realize the desired functional thickness differential (i.e., extra padded thickness **114**) for the area of correction, accommodation, or wedging. The sock orthotic may also have traditional elastic yarn utilized in its fabrication to provide comfort and reduce stretching.

(41) In an embodiment, sole **120** may be marked to indicate the location of the therapeutic area. In an embodiment, sole **120** may be marked to indicate the proper location the metatarsal heads at the ball of the foot, to ensure proper positioning and fit of the sock on the foot. Such marking may be visible on the bottom surface of sole **120**, and the top surface of sock upper **110**.

(42) In an embodiment, if additional pattern padding is needed, a sock orthotic may be manufactured in two jointly attached “mirror” pattern lengths so that when one half is inverted on itself, two identical areas of pattern padding are superimposed, creating a two layer sock with double layers of configured corrections.

(43) In an embodiment, a single sock orthotic may be reversible for use on either left or right foot. In such an embodiment, extra padded section **114** would be evenly distributed above and below base sole section **104**.

(44) The previous description is provided to enable any person skilled in the art to practice the various aspects described herein. In the embodiments, the separation of various system components in the embodiments described above should not be understood as requiring such separation in all embodiments. Various modifications to these aspects will be readily apparent to those skilled in the art, and the generic principles defined herein may be applied to other aspects. Thus, the claims are not intended to be limited to the aspects shown herein, but are to be accorded the full scope consistent with the language claims, wherein reference to an element in the singular is not intended to mean “one and only one” unless specifically so stated, but rather “one or more.” Unless specifically stated otherwise, the term “some” refers to one or more. Pronouns in the masculine (e.g., his) include the feminine and neuter gender (e.g., her and its) and vice versa. Headings and subheadings, if any, are used for convenience only and do not limit the subject disclosure.

(45) A phrase such as an “aspect” does not imply that such aspect is essential to the subject technology or that such aspect applies to all configurations of the subject technology. A disclosure relating to an aspect may apply to all configurations, or one or more configurations. A phrase such as an aspect may refer to one or more aspects and vice versa. A phrase such as a “configuration” does not imply that such configuration is essential to the subject technology or that such configuration applies to all configurations of the subject technology. A disclosure relating to a configuration may apply to all configurations, or one or more configurations. A phrase such as a configuration may refer to one or more configurations and vice versa.

(46) All structural and functional equivalents to the elements of the various aspects described throughout this disclosure that are known or later come to be known to those of ordinary skill in the art are expressly incorporated herein by reference and are intended to be encompassed by the claims.

Claims

1. A sock orthotic comprising a knitted sock having: a sock upper section; and a sock sole section including at least one knitted pattern, wherein: the at least one knitted pattern is an intrinsic part of the sock sole section; at least part of the at least one knitted pattern has a first thickness that is greater than a second thickness of an adjacent part of the sock sole section, thereby forming a

padded region within the sock sole section, where the first thickness is at least $\frac{1}{8}$ " greater than the second thickness when the pad and the adjacent part of the sock sole section are compressed by a user's weight, and the first thickness increases in thickness when uncompressed; and the at least one knitted pattern has a three-dimensional shape and is located on a transverse plane of the sock sole section such that, when the sock orthotic is worn by a user between a foot of the user and a shoe or a weight-bearing surface, the at least one knitted pattern creates an interface between the foot and the shoe or weight-bearing surface, the interface corrective of a biomechanically abnormal area of the foot of the user by having the at least one knitted pattern first thickness at a location on the transverse plane of the sock sole section that is spaced apart from and not directly underneath the biomechanically abnormal area and not applying elastic compression to the biomechanically abnormal area, the at least one knitted pattern selected from the group consisting of an intrinsically knit varus wedge pattern, an intrinsically knit valgus wedge pattern, an intrinsically knit cobra pad; an intrinsically knit metatarsal pad configured to be located immediately proximal to the metatarsal heads and extending proximally to a mid-arch area of the foot; an intrinsically knit neuroma pad; an intrinsically knit dancer's pad; an intrinsically knit heel spur pad, an intrinsically knit heel cut-out perimeter, an intrinsically knit toe crest, an intrinsically knit Morton's extension, an intrinsically knit reverse Morton's extension, and an intrinsically knit Carlton saddle.

2. The sock orthotic of claim 1, wherein the knitted pattern is formed by pulling in additional loops of yarn.

3. The sock orthotic of claim 2, wherein the additional loops of yarn are different from yarn in the sole in at least one of: length, thickness, gauge, elasticity, or density.

4. The sock orthotic of claim 1, wherein a transition from the surrounding sock thickness to the knitted pattern is a step transition.

5. The sock orthotic of claim 1, wherein a transition from the surrounding sock thickness to the knitted pattern is a ramp transition.

6. A method for treating a biomechanically abnormal area of a foot of a user, the method comprising: obtaining, by the user, a knitted sock orthotic comprising a sock upper section; and a sock sole section including at least one knitted pattern, wherein: the at least one knitted pattern is an intrinsic part of the sock sole section; at least part of the at least one knitted pattern has a first thickness that is greater than a second thickness of an adjacent part of the sock sole section, thereby forming a padded region within the sock sole section; where the first thickness is at least $\frac{1}{8}$ " greater than the second thickness when the pad and the adjacent part of the sock sole section are compressed by a user's weight, and the first thickness increases in thickness when uncompressed; and the at least one knitted pattern has a three-dimensional shape and is located on a transverse plane of the sock sole section such that, when the knitted sock is worn by the user between the foot of the user and a shoe or a weight-bearing surface, the at least one knitted pattern creates an interface between the foot and the shoe or the weight-bearing surface, the interface corrective of the biomechanically abnormal area of the foot of the user by having the at least one knitted pattern first thickness at a location on the transverse plane of the sock sole section that is spaced apart from and not directly underneath the biomechanically abnormal area and not applying elastic compression to the biomechanically abnormal area, the at least one knitted pattern selected from the group consisting of an intrinsically knit varus wedge pattern, an intrinsically knit valgus wedge pattern, an intrinsically knit cobra pad; an intrinsically knit metatarsal pad configured to be located immediately proximal to the metatarsal heads and extending proximally to a mid-arch area of the foot; an intrinsically knit neuroma pad; an intrinsically knit dancer's pad; an intrinsically knit heel spur pad, an intrinsically knit heel cut-out perimeter, an intrinsically knit Morton's extension, an intrinsically knit reverse Morton's extension, and an intrinsically knit Carlton saddle; and wearing, by the user, the knitted sock orthotic having a knitted pattern selected from the group to treat the biomechanically abnormal area of the foot.

7. The method of claim 6, wherein the knitted pattern is formed by pulling in additional loops of

yarn.

8. The method of claim 7, wherein the additional loops of yarn are different from yarn in the sole in at least one of: length, thickness, gauge, elasticity, or density.

9. The method of claim 6, wherein a transition from the surrounding sock thickness to the knitted pattern is a step transition.

10. The method of claim 6, wherein a transition from the surrounding sock thickness to the knitted pattern is a ramp transition.
